

Q9.

The screenshot shows the IntelliJ IDEA IDE with a project named "Project Alt+1". The main editor displays a Java file named "q7.java" with the following code:

```
3 class Processor { no usages
9 void process() { no usages
15 }
16 }
17
18 public class q7 {
19
20 public static void main(String[] args) {
21     try {
22         int a = 10, b = 0;
23         int result = a / b;
24         System.out.println(result);
25     }
26     catch (ArithmeticException e) {
27         System.out.println("Arithmetic error: " + e);
28     }
29     catch (Exception e) {
30         System.out.println("General error: " + e);
31     }
32
33     try {
```

Below the code editor, the "Run" tab is active, showing the execution output:

```
Run q7 x
C:\Users\sabar\.jdk\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\lib\idea_r
Arithmetic error: java.lang.ArithmeticException: / by zero
Array index error: java.lang.ArrayIndexOutOfBoundsException: Index 5 out of bounds for length 3
Done
```

Q.8.

```
1 public class Main{
2     public static void main(String[] args){
3         final int speed;
4         speed = 100;
5         System.out.println(speed);
6         System.out.println(Constants.MAX_SPEED);
7     }
8 }
9 final class Constants{
10     static final int MAX_SPEED = 120;
11 }
12 class SuperClass{
13     public final void show(){
14         System.out.println("This isa final keyword.");
15     }
16 }
17 class SubClass extends SuperClass{
18
19 }
20 class AnotherClass{
21
```

OUTPUT

```
100
120
```

Q.7.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main{
2     public static void main(String[] args)throws InterruptedException{
3         MyThread t1 = new MyThread();
4         t1.start();
5         Thread.sleep(100);
6         System.out.println("Main thread Finished.");
7     }
8 }
9 class MyThread extends Thread{
10     public void run(){
11         System.out.println("Thread running");
12     }
13 }
14 class Counter implements Runnable{
15     private int count = 0;
16
17     public void increment(){
18         count++;
19     }
20     public void run(){
21         increment();
```

OUTPUT

Thread running
Main thread Finished.